



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
14-03-2021

JEE(Main + Advanced) : NURTURE PHASE-I, I(B), NURTURE LIVE PHASE-I & I(B)

PAPER-1

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

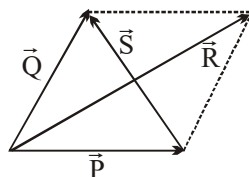
SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

-
1. Which one of the following physical quantity has different dimension than the rest?
(A) Stress (B) Torque (C) Work (D) Energy
 2. If instead of mass, length and time as fundamental quantities, we choose velocity, acceleration and force as fundamental quantities and express their dimensions by V, A and F respectively, then the dimensions of Young's modulus 'Y' will be as expressed as : Given $[Y] = [ML^{-1}T^{-2}]$
(A) $[FA^2V^{-4}]$ (B) $[F^2V^{-1}A]$ (C) $[FA^2V^{-1}]$ (D) $[FAV^{-2}]$
 3. In a new unit system, 1 unit of time is equal to 20 second, 1 unit of mass is 15 kg and 1 unit of length is 20 m. In this new system of units, 1 unit of energy is equal to
(A) 15 Joule (B) $\frac{1}{20}$ Joule (C) 4 Joule (D) 16 Joule
-

Space for Rough Work

4. Two vectors $\vec{P}$ & $\vec{Q}$ are arranged in such a way that they form adjacent sides of a parallelogram as shown in figure. Which of the following options have incorrect relationship :-



- (A) $\vec{Q} = \vec{P} + \vec{S}$ (B) $\vec{R} = \vec{P} + \vec{Q}$ (C) $\vec{R} = \vec{P} + \vec{S}$ (D) $\vec{S} = \vec{Q} - \vec{P}$
5. Find the component of the vector $2\hat{i} + 3\hat{j} + \hat{k}$ along the z-x plane.
- (A) $\sqrt{14}$ (B) $\sqrt{5}$ (C) $\sqrt{10}$ (D) 3
6. A bird moves from point (1, -2) to (4, 2). If the speed of the bird is 20 m/s, then the velocity vector of the bird is
- (A) $5(\hat{i} - 2\hat{j})$ (B) $6\hat{i} + 8\hat{j}$ (C) $0.6\hat{i} + 0.8\hat{j}$ (D) $12\hat{i} + 16\hat{j}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. A dimensionless quantity y is represented by formulae $y = \frac{p - q \cdot r}{s + t}$, which of the following is(are) correct ?

- (A) Dimension of t and s are same. (B) $\frac{qr}{p + s}$ is dimensionless.
(C) pqr and st have same dimensions. (D) $st + qr$ is not meaningful.

Space for Rough Work

8. Which of the following relations cannot be derived from dimensional analysis :-

(A) $s = ut + \frac{1}{2}at^2$

(B) Volume = Area $\times$ Height

(C) Force = $k \times \text{mass} \times \text{acceleration}$

(D) Force = $\eta A \frac{\Delta v}{\Delta x}$

where s is displacement, u – initial velocity, t – time, a – acceleration, k – dimensionless constant, η – coefficient of viscosity, A – Area, V – velocity, x – distance.

9. If angle between $\vec{A}$ and $\vec{B}$ is 90° and both vectors have same magnitude :

(A) $\vec{A} \cdot \vec{B} = 0$

(B) $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$

(C) $\vec{A} + \vec{B} = \vec{A} - \vec{B}$

(D) $(\vec{A} + \vec{B}) \cdot (\vec{A} - \vec{B}) = 0$

10. If $\vec{A} = \hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} + \hat{k}$ then choose the correct option(s) :

(A) $\vec{A}$ and $\vec{B}$ are perpendicular to each other.

(B) $|\vec{A}| = |\vec{B}|$

(C) $\vec{A}$ and $\vec{B}$ are parallel vectors.

(D) $\vec{A} + \vec{B}$ is perpendicular to $\vec{A} - \vec{B}$.

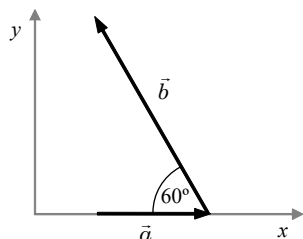
Space for Rough Work

11. Which of the following vector(s) has projection of length '5' on xy-plane :-

- (A) $3\hat{i} + 4\hat{j} + 3\hat{k}$ (B) $4\hat{i} - 3\hat{j} + 3\hat{k}$ (C) $5\hat{i} + 7\hat{k}$ (D) $5\hat{i} + 4\hat{j} + 3\hat{k}$

12. Two vectors $\vec{a}$ and $\vec{b}$ of magnitudes 2 units and 4 units respectively are shown in the figure.

Which of the following mathematical operations is/are correct?



- (A) $\vec{a} + \vec{b} = 2\sqrt{3}\hat{j}$ (B) $\vec{a} - \vec{b} = 4\hat{i} - 2\sqrt{3}\hat{j}$ (C) $\vec{a} \cdot \vec{b} = 4$ (D) $(\vec{b} \cdot \hat{a}) = 2$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Power transmitted by wave in a string is given by $P = kv \mu^a \omega^b A^c$; where k is dimensionless constant

v = velocity of wave

μ = mass per unit length of string

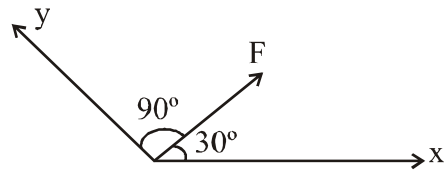
ω = angular frequency of wave $\left(\omega = \frac{\theta}{t} \right)$

A = Amplitude of wave (it is the maximum distance of particle from its mean position)

Then, the value of $ab + c^2$ is

Space for Rough Work

2. Two forces of magnitudes P & Q are inclined at an angle (θ) the magnitude of their resultant is $5Q$. When Q is reversed in direction, the magnitude of the resultant of the forces becomes Q. Find the value of $\frac{P}{Q\sqrt{3}}$.
3. Two vectors $\vec{A} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ and $\vec{B} = 4\hat{i} + 8\hat{j} + x\hat{k}$ are such that the components of $\vec{B}$ along $\vec{A}$ is zero. Then the value of x is :
4. x and y are axes. F is a vector as shown in the figure. The ratio of x and y component of vector F is :



Space for Rough Work

5. if dot product of $\vec{A} = 2\hat{i} + 3\hat{j}$ with a vector $\vec{B}$ which is in y-z plane comes out to be 15 then magnitude of component of $\vec{B}$ in x-y plane is :
6. A vector $\mathbf{a}$ of magnitude 8 units has two components. One is perpendicular to $\mathbf{a}$ and is of magnitude 6 units. What is the magnitude of the other component?
[Hint: They are 'components' not 'rectangular component']

Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks : 18)**

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Negative Marks : -1 In all other cases

1. Which of the following element will have highest number of exchanges?
(A) Carbon (B) Nitrogen (C) Oxygen (D) Fluorine
2. Which of the following element belongs to d-block and is not a transition element?
(A) La (B) Zn (C) Ac (D) Hf
3. If x are the number of elements in 4th period of periodic table, then number of elements in 7th and 9th period would be respectively :-
(A) $\frac{16}{9}x, \frac{16}{9}x$ (B) $\frac{25}{9}x, \frac{25}{9}x$ (C) x, x (D) $\frac{16}{9}x, \frac{25}{9}x$

Space for Rough Work

4. Select the incorrect size order.
(A) $\text{Cl} > \text{Cl}^+ < \text{Cl}^-$ (B) $\text{Li} < \text{Ne} > \text{F}$ (C) $\text{Li}^+ < \text{Mg}^{2+} > \text{Be}^{2+}$ (D) $\text{Na}^+ < \text{K}^+ > \text{Li}^+$
5. Which pairs is in the order of decreasing value of Z_{eff} ?
(A) (Na, K, Rb) (B) (P, S, Cl) (C) (Mg, Ca, Sr) (D) (Be, He, Li)
6. Less electropositive elements occupied _____ part of Lothar Meyer curve.
(A) Ascending (B) Descending (C) Bottom (D) Peak

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

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-
7. Select the option(s) which are true for electron filling order?
- (A) First, less energy shell is filled irrespective of subshell in multi electronic species
- (B) While filling electrons in degenerate orbitals, firstly each orbital is filled with same spin electrons
- (C) Electrons are removed from subshells in exactly the same sequence they are filled
- (D) Unpaired electron presence provides certain magnetic nature to metal atom/ion

Space for Rough Work

8. Consider the following electronic configuration
 $X \Rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$
 Select the incorrect options.
 (A) No rules are violated in the case of given electronic configuration
 (B) X belongs to coinage metal group
 (C) X is only element having exceptional electronic configuration in its period and group
 (D) If the configuration would have been $[Ar]3d^9 4s^2$, number of exchanges would remain same.
9. Which of the following element(s) are both typical and bridging elements?
 (A) Cl (B) P (C) Mg (D) S
10. The exothermic processes from the following are :-
 (A) $Ne^- \longrightarrow Ne$ (B) $F \longrightarrow F^-$ (C) $Cl \longrightarrow Cl^+$ (D) $O^- \longrightarrow O^{-2}$
11. Select the d-block element(s).
 (A) $Hf \Rightarrow [Kr] 4f^{14} 5d^2 6s^2$
 (B) $Ce \Rightarrow [Kr] 4f^1 5d^1 6s^2$
 (C) $Th \Rightarrow [Xe] 6d^2 7s^2$
 (D) $Ac \Rightarrow [Xe] 6d^1 7s^2$
12. For F^- ,
 (A) $\sigma = 0.35 \times 7 + 2 \times 0.30$
 (B) $\sigma = 0.35 \times 7 + 2 \times 0.85$
 (C) $Z_{eff} = 10 - (0.35 \times 7 + 2 \times 0.30)$
 (D) $Z_{eff} = 9 - (0.35 \times 7 + 2 \times 0.85)$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

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+		-	
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

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1. $\frac{\text{Number of p-electrons in Ar}}{\text{Number of s-electrons in Be}} = ?$

2. (C, O), (N, P), (S, Se), (Ne, F), (H, He)

Number of pairs where first element have higher EA than second are :-

3. For Cr,

group number = x

period number = y

number of valence electrons = z

then $\frac{x}{y+z} = ?$

Space for Rough Work

4. Fe^{2+} , Mn^{2+} , Co^{2+} , Ni^{2+} , V^{3+} , Sc^{+3} , Ti^{4+}

Number of ions having more than 2 unpaired electrons are = x

Number of diamagnetic ions = y

find $\frac{x}{y} = ?$

5. Z_{eff} of Na $\times$ Z_{eff} of Rb = ?

6. Maximum number of electrons with set of quantum numbers $n = 5$, $m = 0$
are = ?

Space for Rough Work

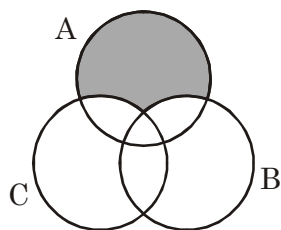
PART-3 : MATHEMATICS
SECTION-I(i) : (Maximum Marks : 18)

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-
1. Number of integers satisfying the inequality $\frac{3x+8}{4x-21} \leq 0$ is
(A) 7 (B) 9 (C) 8 (D) 10
2. If $x > 0$, then the minimum value of $\frac{x}{3} + \frac{27}{x}$ is
(A) 4 (B) 3 (C) 6 (D) 9
3. If $2x^3 - x^2 - 22x + \lambda = 0$ has one root as 4 then sum of other two roots is
(A) -7 (B) -3.5 (C) 3.5 (D) 7
-

Space for Rough Work

4. The shaded region in the given figure is



- (A) $A \cap (B \cup C)$ (B) $A \cup (B \cap C)$ (C) $A \cap (B - C)$ (D) $A - (B \cup C)$
5. If $\frac{3\sqrt{2}}{\sqrt{6}+\sqrt{3}} - \frac{4\sqrt{3}}{\sqrt{6}+\sqrt{2}} + \frac{\sqrt{6}}{\sqrt{3}+\sqrt{2}} + \frac{1}{\sqrt{5}+\sqrt{21}} = \sqrt{\frac{a}{8}} - \sqrt{\frac{b}{8}}$ then the value of $a + b$ is equal to
- (A) 8 (B) 9 (C) 10 (D) 11
6. If $A = \{x : (x - 3)(x + 2) \leq 0 \text{ \& } x \in \mathbb{N}\}$, then the number of proper subsets of set A is
- (A) 61 (B) 15 (C) 62 (D) 7

Space for Rough Work

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7. If $\left(\frac{x^{2/3} \cdot y^{-1/2}}{y \cdot x^{-2/3}} \cdot \sqrt{\frac{x \cdot y^{-2}}{y \cdot x^{-1}}} \right)^9 = x^{k_1} \cdot y^{k_2}$, then which of the following is/are **TRUE** ?

- (A) $k_1 - k_2 = 48$ (B) $k_1 + k_2 = 48$ (C) $k_1 - k_2 = -6$ (D) $k_1 + k_2 = -6$

8. If $P(x)$ is a cubic polynomial such that $P(0) = 1$, $P(1) = 4$, $P(2) = 9$ with leading coefficient 2 then

- (A) $P(-1) = -12$ (B) $P(-1) = 12$ (C) $P(3) = 28$ (D) $P(3) = 25$

Space for Rough Work

9. If $\left(\sqrt{3+2\sqrt{2}}\right)^{3x} + \left(\sqrt{3-2\sqrt{2}}\right)^{3x} = 2\sqrt{2}$, then which of the following is/are hold(s) good ?

- (A) Number of solutions of x is 2 (B) Number of solution of x is 1
(C) Sum of solutions is 0 (D) Sum of solutions is 1/3

10. The value(s) of x satisfying the equation $x^2 + \frac{1}{x^2} - 10\left(x + \frac{1}{x}\right) + 23 = 0$ is

- (A) $\frac{7+\sqrt{45}}{2}$ (B) $\frac{3+\sqrt{5}}{2}$ (C) $\frac{3-\sqrt{5}}{2}$ (D) $\frac{7-\sqrt{45}}{2}$

11. Sum of all roots (real or non real) of the equation $(x+1)(x+2)^2(x+4) - 2x^2 = 0$ is greater than

- (A) -9 (B) -10 (C) -11 (D) -8

12. If $\frac{p}{q} = \frac{x+1}{x-1}$ and $f(x) = \frac{p^2 - q^2}{p^2 + q^2}$ then identify the correct statement(s)

- (A) $f(1) = 1$ (B) $f(1) = -1$ (C) $f(0) = 0$ (D) $f(0) = 2$

Space for Rough Work

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+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Product of real roots of the equation $(x + 1)(x + 2)^2(x + 4) - 2x^2 = 0$ is
- If $x, y, z \in \mathbb{R}$ and $49x^2 + 4y^2 + 9z^2 - 70x - 4y - 6z + 27 = 0$, then the value of $x + y + 3z$ is
- If $x = \sqrt{2 - \sqrt{3}}$ and $y = \sqrt{2 + \sqrt{3}}$. If the value of the expression $x - y + 2x^2y + 2xy^2 - x^4y + xy^4$ can be expressed in the form $\sqrt{p} + \sqrt{q}$ ($p, q \in \mathbb{N}$), then the value of $p + q$ is

Space for Rough Work

4. If x & y are positive integers, such that $\frac{1}{x} + \frac{1}{y} = \frac{1}{7}$ & $x \geq y$, then the number of ordered pairs of (x, y) is
5. If $(5 + 1)(5^2 + 1)(5^4 + 1)(5^8 + 1) \dots (5^n + 1) + \frac{1}{4} = \frac{5^{n^2+4n-288}}{4}$, then the value of $\frac{n}{4}$ is
($n = 2^p$ where $p \in \mathbb{N}$)
6. If the expression $x^4 + 3x^2y^2 + 4y^4$ is factorised as $(ax^2 + by^2 - axy)(x^2 + by^2 + axy)$, then the value of $\frac{a}{b}$ is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	18
I(ii)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator